

PROJECT REPORT

Old Car Engine Health Diagnostic via Audio

(AI & Machine Learning Based Mini Project)

Submitted by

Sparsh Jain

Subject

Fundamental of AI & ML

Project Type

AI & ML Mini Project

Topic

Old Car Engine Health Diagnostic via Audio

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This report explains a simple AI and machine learning idea that uses the sound of an old car engine to give an early indication of its condition. The language has been kept simple and student-friendly so the report reads like a normal project explanation.

1. Abstract

The Old Car Engine Health Diagnostic via Audio project is a small AI and machine learning based system that tries to identify whether an old car engine sounds normal or may have a problem. Instead of checking only by looking at the engine, the system uses an audio recording of the engine while it is running. The sound is then converted into useful numerical features and given to a machine learning model.

The main idea is simple: different engine conditions can produce different sound patterns. A healthy engine generally has a more regular sound, while problems such as unusual knocking, rough running, misfiring or abnormal vibration can change the audio pattern. The model learns these patterns from labelled engine sound examples and predicts the condition of a new recording.

The project is mainly made to understand basic AI and ML concepts such as data collection, feature extraction, training data, testing data, classification and prediction. It is not intended to replace a professional mechanic. It can be treated as an early-warning or educational tool.

2. Introduction

2.1 Background

When an old car starts making a different sound, the owner may not immediately know what the sound means. A mechanic can listen to the engine and inspect the vehicle, but this may not always be possible at the moment. Audio-based analysis gives an interesting way to use AI for an everyday problem.

In this project, the sound of a running engine is recorded using a microphone or mobile phone. The audio is processed so that the computer can work with it. Features such as frequency information, energy and changes in the signal can then be used by a machine learning model to learn the difference between normal and abnormal engine sounds.

2.2 Objectives

- Create a simple AI/ML concept for checking old car engine sounds.
- Use engine audio as the main input instead of only manual inspection.
- Convert raw audio into numerical features that a machine learning model can understand.
- Train a classification model using labelled engine sound samples.
- Predict whether a new engine recording appears normal or shows an abnormal pattern.
- Understand the basic workflow of an AI and ML project.

3. System Analysis

3.1 Existing Method

- Manual listening: A mechanic listens to the engine and uses experience to identify unusual sounds.
- Physical inspection: Parts of the engine are checked directly for visible or mechanical problems.
- Scanner-based diagnosis: Modern vehicles can also provide electronic fault codes, but older cars may have limited electronic diagnostics.
- Human judgement: The result can depend on the experience, environment and listening conditions of the person checking the engine.

3.2 Proposed System

The proposed system uses AI and machine learning to analyse engine audio. The user records a short sound clip of the car engine. The program cleans or prepares the audio, extracts important features and sends them to a trained ML model. The model then gives a simple output such as Normal Engine Sound or Possible Abnormal Sound.

The system is designed as a basic learning project. A prediction does not prove that a mechanical fault exists. It only shows whether the recorded sound is similar to the patterns present in the training data.

Main benefits:

- Uses a low-cost input such as a phone or microphone.
- Can process an audio recording without opening the engine.
- Demonstrates a real-life use of AI and machine learning.
- Can be improved by adding more labelled engine sound data.
- Produces a quick classification for learning and early screening.

4. System Design

4.1 Basic Structure

- Input: Audio recording of an old car engine.
- Pre-processing: Remove unwanted noise and prepare the audio signal.
- Feature Extraction: Calculate useful audio features such as MFCC, spectral information and signal energy.
- Machine Learning: A trained classifier learns the relationship between audio features and engine condition.
- Output: Predicted class such as Normal or Possible Abnormal.

4.2 How the Data Moves Through the System

1. The engine audio is recorded.
2. The audio file is loaded into the program.
3. Basic pre-processing is applied.
4. Audio features are extracted.
5. The trained ML model receives the feature values.
6. The model predicts the most suitable class.

7. The result is displayed in simple language.

4.3 Main AI/ML Concepts

Concept	Simple Explanation
Training Data	Recorded engine sounds with known labels used to teach the model.
Features	Numerical information taken from the audio signal.
Classification	Putting an audio sample into a condition category.
Model	A machine learning method that learns patterns from training examples.
Testing Data	Separate examples used to check how the trained model performs.
Prediction	The class produced by the model for a new engine recording.

4.4 Audio Features

Raw audio contains a large number of signal values. A machine learning model usually works better when useful information is extracted from the signal. Common features for this type of project can include MFCCs, zero-crossing rate, spectral centroid, spectral bandwidth and RMS energy. These features describe different parts of the engine sound in numerical form.

MFCC (Mel-Frequency Cepstral Coefficients) is especially useful in audio classification because it represents important characteristics of a sound in a compact numerical form. In this project, MFCC values can be combined with other features to create the input for the ML model.

4.5 Possible Model

For a beginner-level project, a classification algorithm such as Random Forest, Support Vector Machine or Logistic Regression can be used. Random Forest is easy to understand because it combines the decisions of multiple decision trees. The exact model can be changed later depending on the amount and quality of available data.

4.6 Simple Classification Formula

The model receives an input feature vector X extracted from an engine recording and produces a predicted class Y . In simple terms:

Engine Audio → Features → ML Model → Predicted Engine Condition

5. Implementation Details

5.1 Hardware and Software Requirements

Requirement	Details
Operating System	Windows / macOS / Linux
Programming Language	Python
Python Version	Python 3.x
Libraries	NumPy, Pandas, Librosa, Scikit-learn, Matplotlib
Editor / IDE	VS Code / PyCharm / Jupyter Notebook
RAM	4 GB or more recommended
Input Device	Mobile phone microphone or external microphone

5.2 Python / ML Concepts Used

- NumPy – handles numerical values and arrays.
- Pandas – helps organise labelled audio feature data.
- Librosa – loads audio and extracts useful sound features.
- Scikit-learn – provides machine learning algorithms and evaluation tools.
- Train-test split – separates training examples from testing examples.
- Random Forest / SVM – can be used for engine sound classification.
- Accuracy and confusion matrix – help understand model performance.

5.3 Main Functions

Function	Purpose
load_audio()	Loads the recorded engine sound.
extract_features()	Converts the audio into numerical features.
prepare_dataset()	Combines features with their engine-condition labels.
train_model()	Trains the selected machine learning classifier.
predict_condition()	Predicts the condition of a new audio recording.
main()	Runs the complete process and displays the result.

5.4 Sample Input

Field	Sample Value
Audio File	old_car_engine_01.wav
Recording Length	10 seconds

Field	Sample Value
Engine State	Normal
Sample Rate	22,050 Hz
Environment	Outdoor / low background noise

5.5 Sample Dataset Structure

A small labelled dataset can be prepared by recording engines in different conditions. Each audio file is given a label. For example, a basic dataset may contain normal engine sounds and sounds recorded when an unusual noise is present.

Audio File	Label	Example Description
engine_01.wav	Normal	Regular engine sound
engine_02.wav	Normal	Regular idle sound
engine_03.wav	Abnormal	Unusual knocking pattern
engine_04.wav	Abnormal	Rough or irregular sound
engine_05.wav	Normal	Stable engine sound

6. Features and Functionality

6.1 Audio Input

The system accepts a recorded engine sound as its input. A short recording is enough for a basic demonstration, although real-world performance depends heavily on recording quality, microphone position and the type of engine sounds included in the training data.

6.2 Feature Extraction

The program converts the sound into numerical values. These values represent characteristics of the audio and become the input for the machine learning model.

6.3 Machine Learning Prediction

After training, the model receives features from a new audio file. It compares the learned patterns and returns the class that best matches the input according to the trained model.

6.4 Human-Language Result

Instead of showing only a technical class number, the program can display a simple message such as: **“The engine sound appears similar to the normal samples.”** or **“The engine sound has an abnormal pattern. Further mechanical checking is recommended.”**

These messages are only screening results. The system should not claim that a specific mechanical part is definitely damaged unless there is sufficient labelled data and proper validation for that fault.

7. Testing and Expected Output

The model can be tested using audio files that were not used during training. This is important because a model that performs well only on its training data may not work well on new recordings.

Test	Input	Expected Output
Test 1	Normal engine recording	Normal
Test 2	Abnormal engine recording	Possible Abnormal
Test 3	Noisy recording	Prediction may be less reliable
Test 4	Different engine type	Performance may vary

7.1 Evaluation

For a basic classification project, accuracy can be used as one evaluation measure. Precision, recall and a confusion matrix can also be checked when the dataset is large enough. The evaluation should be performed on separate test data so that the result gives a better idea of how the model handles unseen recordings.

Example:

Metric	Example Value
Training Samples	80
Testing Samples	20
Correct Predictions	17
Incorrect Predictions	3
Example Test Accuracy	85%

The values above are only sample project values to explain testing. They are not a claim about the accuracy of a real engine-diagnostic system. Actual performance would need to be measured using a properly collected and labelled dataset.

8. Advantages and Limitations

8.1 Advantages

- Uses a simple and inexpensive audio input.
- Shows a practical application of AI and machine learning.
- Can provide a quick first-level screening result.
- The system can be improved by adding more engine sound recordings.
- The output can be written in simple human language.

8.2 Limitations

- Background noise can affect the audio features and prediction.
- Different car models and engine types can produce different normal sounds.

- A small or poorly labelled dataset can give unreliable predictions.
- The model cannot replace a professional mechanical inspection.
- A predicted abnormal sound does not identify the exact damaged part.
- Real-world use would require much more testing and validation.

9. Future Scope

- Larger Dataset: Record more cars, engine types and fault conditions.
- Mobile Application: Build an app that records the engine sound and gives the result directly.
- Better Noise Handling: Add stronger audio cleaning and noise-reduction methods.
- Multi-Class Diagnosis: Train the model for different sound categories such as normal, knocking, misfire and belt-related noise, if enough labelled data is available.
- Deep Learning: Try CNN or other audio-classification models after building a sufficiently large dataset.
- Real-Time Analysis: Analyse engine sound continuously while the vehicle is running.
- Confidence Display: Show a model confidence value together with a clear warning that it is only an AI-based screening result.
- Maintenance History: Combine repeated audio checks with dates and vehicle information.

10. Conclusion

This project shows how the basic ideas of Artificial Intelligence and Machine Learning can be used with a common real-life problem. Instead of depending only on manual listening, the system converts an old car engine sound into numerical audio features and uses a machine learning model to identify patterns.

The project also makes it clear that the quality of the data is very important. A model can only learn useful patterns when the training recordings are varied, correctly labelled and recorded in suitable conditions. Because engine sounds can be affected by the car model, microphone, background noise and recording position, the result should be treated as an early screening indication rather than a final mechanical diagnosis.

Overall, the project is useful for understanding the basic AI/ML workflow: collect data, prepare it, extract features, train a model, test it and use the trained model to make a prediction. It provides a simple starting point for developing a more advanced vehicle sound diagnostic system in the future.

11. References

1. Librosa documentation – audio analysis and feature extraction.
2. Scikit-learn documentation – machine learning and model evaluation.
3. NumPy documentation – numerical computing in Python.
4. General concepts of audio signal processing and machine learning classification.
5. Project design and example dataset structure prepared for the Old Car Engine Health Diagnostic via Audio project.

Project Deliverable

The project can be implemented as a Python-based AI/ML program. A typical deliverable may contain the Python source code, labelled engine audio dataset, extracted feature file, trained model file and this project report.